WHO'S AFRAID OF THE MINIMUM WAGE? MEASURING THE IMPACTS ON INDEPENDENT BUSINESSES USING MATCHED U.S. TAX RETURNS*

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A common concern surrounding minimum wage policies is their impact on independent businesses, which are often feared to be less able to bear or pass on cost increases. We examine how these typically small and medium-size firms accommodate minimum wage increases along product and labor market margins using a matched owner-firm-worker panel data set drawn from the universe of U.S. tax records over a 10-year period, and using state minimum wage changes as identifying variation. We find that on average, firms in highly exposed industries do not substantially reduce employment—they do not lay off workers but moderately reduce part-time hiring. Instead, these firms are able to fully finance the new labor costs with new revenues, leaving average owner profits unchanged. Higher wage floors, however, forestall entry, particularly for less productive firms, reducing the number of independent firms operating in these industries by roughly 2%. Yet these industries do not shrink; instead, incumbent responses and strong positive selection among entrants reshape industries that rely heavily on low-wage workers, yielding fewer but more productive firms after the cost shock. We also take a worker-level perspective to examine how potentially vulnerable individuals are affected by minimum wage increases. Using panels of low-earning and

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young workers, we find that their average earnings rise substantially with the minimum wage, while they are no less likely to be employed. Worker transitions indicate that minimum wage increases boost retention and that worker reallocation from independent firms toward corporations buffers disemployment impacts from reduced hiring at independent firms. *JEL codes:* J23, J31, J38, L13, L11.

I. Introduction

Proposals to raise the minimum wage are often met with arguments that independent businesses may be particularly vulnerable to wage-floor increases.[1] Although recent research has found that minimum wage increases have had few deleterious aggregate employment effects in the short run,[2] fears that independent businesses operate on margins too slim to accommodate cost increases or face demand too elastic to pass costs through to consumers have motivated small business exemptions and even wholesale opposition to raising wage floors. At the same time, surveys repeatedly find independent business owners divided on minimum wage policy, with large shares actually supporting higher wage floors.[3]

This article presents the first comprehensive examination of how independent firms accommodate minimum wage increases in the United States. To track independent firms, we construct a novel linked firm-worker-owner panel data set from the universe of U.S. tax returns. To measure responses in both product and labor markets, we match the tax returns of pass-through firms—the predominant organizational form of independent businesses—to the individual income tax returns of their workers and owners over a 10-year period. These data allow us to measure firm responses across a number of margins, including employment, compensation to different workers, expenditures on nonlabor inputs,

1. For example, see articles by Josephs and Janofsky (2015) in the Wall Street Journal and Dunkelberg (2021) in Forbes.

2. See Cengiz et al. (2019), Dustmann et al. (2022), Dube, Lester, and Reich (2010), Allegretto and Reich (2018), and Harasztosi and Lindner (2019), plus Belman and Wolfson (2014) and Clemens (2021) for reviews of recent research.

3. For example, see polls from Gallup (Dugan 2013), CNBC (Wronski and Cohen 2020), the American Sustainable Business Council (ASBC and BFMW 2014), the Society for Human Resource Management (Mayer 2023), and reporting in the Washington Post (DePillis 2016).

revenues, profits, and firm exit and entry. We complement the firm-level analysis with an individual-level analysis that follows low-earning and young individuals over time and across employers to understand how minimum wage increases affect those most likely to be affected by wage-floor policies.

To estimate the causal effects of minimum wage increases, we exploit subnational state policy changes, allowing us to compare treated firms to similar but wholly untreated firms in states that did not adjust their wage regulations, controlling flexibly for firm size, industry, and market characteristics. Our long panel helps validate the causal interpretation of the findings by providing several years of data to assess pre-trends in treatment and control states and to conduct placebo tests. The breadth of our data, which describe 271,000 firms, allows us to capture sector-level effects and to measure how responses vary by firm and worker characteristics.

Using these data and a difference-in-differences empirical strategy to compare firms in states that did and did not raise their minimum wage, we examine the various margins along which independent firms accommodate the cost shock associated with the minimum wage increases. First, we examine whether and how firms adjust employment when wage floors rise. We find that firms do not lay off employees but modestly reduce hiring in part-time positions, resulting in 1.5 (or 2%) fewer employment relationships per firm per year. The reduction in hiring is fully concentrated in positions paying less than $4,000 a year, mostly held by teenagers.

Next, we examine how independent firms accommodate higher minimum wages given that they do not substantially reduce employment. Higher minimum wages drive up the wage bills of exposed firms by 6% on average, with income gains concentrated among low-earning workers and no evidence of reduced pay to high-earning workers. By estimating changes in each element of firms' tax returns and net profits, we can determine who bears the economic burden of these labor cost increases. We find that independent firms finance their higher wage bills with new revenue, raising roughly 3.3% more revenue four years after the minimum wage increase. Revenue rises enough to leave profits fully intact, meaning that value added per worker increases sufficiently for consumers to entirely finance the higher pay to low-earning workers, while firm owners bear none of the economic burden.

Minimum wage increases are not a free lunch for firms and their owners. In the years after a state raises its minimum wage, entry slows in highly exposed industries. Four years after a wage-floor increase, independent-firm entry is dampened by 2 percent, meaning that 2 percent fewer independent firms operate in these industries. Firms that do enter are positively selected, featuring higher productivity and greater cost efficiency, but nonetheless dedicate higher shares of revenue to wage bills, much like incumbent firms. Among incumbents, the cost structure changes, leading them to spend proportionately less on nonlabor inputs as they increase sales through greater value added per worker.

The extensive-margin dynamics and responses among incumbent firms combine to shape the cost structure and productivity of these sectors overall. Despite dampened entry, the positive selection of entrants facilitates a net shift in the productivity distribution of active firms. Furthermore, we find that these sectors do not shrink—aggregate revenues across all independent firms in highly exposed industries rise by 2.63% (std. err. = 0.0294) and aggregate profits are unchanged. Total pay to workers increases by 1.1% (std. err. = 0.0061) of baseline revenues; as a result, workers receive a bigger slice of a larger revenue pie. To contextualize our findings, we present a simple conceptual framework of imperfect product market competition with fixed costs to highlight how extensive-margin responses and heterogeneity can mediate observed responses to minimum wage increases.

Although minimum wages only modestly affect employment from a firm perspective, reduced part-time hiring and reduced firm entry could deny young workers a foothold in the labor market and decrease the number of jobs available to low-earning workers. Examining individual panels of low-earning workers and young individuals, we find that their annual earnings rise on average by 18.9% and 21.8%, respectively, in the years after the minimum wage increase. Their employment remains essentially unchanged relative to similar workers in untreated states, with estimated changes of −0.24% (std. err. = 0.0068) for low earners and 1.39% (std. err. = 0.0038) for young workers, corresponding to own-wage employment elasticities of −0.013 (std. err. = 0.036) and 0.064 (std. err. = 0.032). In addition, retention rates rise: four years after the minimum wage increase, we estimate a 1.87% reduction in separation rates among low-earning workers at highly exposed firms at baseline.

Minimum wage increases also alter worker transition patterns, making workers more likely to move to larger firms, especially large C-corporations. These transition patterns help reconcile the null employment effects we estimate in the individual panels with the modest reductions in employment relationships we document among independent firms in highly exposed industries: workers sustain employment in part by moving away from smaller independent firms to larger corporations. Higher retention rates also coincide with reductions in part-time hiring, indicating that reduced turnover partly explains the decline in the number of employment relationships at independent firms that we observe. Overall, we find that on average, minimum wage increases have little effect on employment among potentially vulnerable firms and workers, which complements estimates from Card and Kreuger (1995) and recent studies finding small short- to medium-run employment effects of the minimum wages.

This study makes three key contributions. First, we provide comprehensive documentation of how firms accommodate higher wage floors. Ours is the first study to trace the incidence of minimum wages for a large set of firms in the United States, incorporating product market, labor market, and productivity response margins. We focus on independent firms, which are politically and economically important as their welfare is central to minimum wage policy debates, and collectively they account for roughly half of employment in the United States. Drawing from the universe of administrative tax records, we have a long panel of firms that allows us to provide a detailed analysis of the joint responses to minimum wage increases along various margins, similar to Harasztosi and Lindner (2019).[4] Our firm-worker link allows us to look beyond net employment changes and estimate how employment relationships change across types of work-

4. Previous work has assessed responses across specific margins such as revenue and profits (Draca, Machin, and Reenen 2011), pass-through (Leung 2021; Renkin, Montialoux, and Siegenthaler 2022), employment, worker reallocation (Dustmann et al. 2022), and exit (Luca and Luca 2019). Other work has incorporated multiple response margins, but for establishments of a single firm (Brummund 2018; Ashenfelter and Jurajda 2021; Coviello, Deserranno, and Persico 2022), a set of franchisees (Hirsch, Kaufman, and Zelenska 2015), or a particular industry (Ruffini 2024).

ers. Because our data also comprise annual representative cross sections of independent firms, we can track effects on new firms and therefore measure effects on aggregates. To the deep literature on wage floors, we add fresh evidence that minimum wage policies in the United States mainly redistribute from customers to workers rather than within small firms from owners to workers, with some would-be entrepreneurs also bearing the costs of reduced entry.

Second, our identification strategy, based on state policy changes, allows us to estimate new dimensions of the effects of minimum wage increases. The natural control group of untreated states provides a sector-level counterfactual that is key to measuring how higher wage floors affect firm entry and productivity dynamics.[5] In particular, because our identification does not depend on differences in pre-reform firm exposure, we can estimate how minimum wages affect firm entry and, in turn, how selection among entrants contributes to shifts in the productivity distribution of highly exposed industries. Further, the policy variation we exploit also mirrors proposed incremental minimum wage increases, which primarily affect the service sector.

Third, the breadth and duration of our individual panel data mean that we can precisely evaluate how well minimum wages benefit the people they aim to help. Our precise estimates of the employment and earnings responses even for subgroups of individuals allow us to measure near-zero and small employment effects with tight confidence intervals. Because our data describe total earnings, which encompass any changes in wages, employment, and hours, our estimates capture how minimum wages affect the material well-being of low-income people. Our panels track employment in any industry and firm type, linking workers to their employers before and after the minimum wage increase. The link between workers and employers allows us to contextualize the firm-level results, showing that minimum wage increases boost retention and reallocate low-earning and teen workers from independent businesses to C-corporations. These changes in transition dynamics buffer employment responses and reconcile the firm and individual results.

5. We also side-step recently raised concerns (Haanwinckel 2025) about identification drawing on national variation and using “fraction affected” and “effective minimum wage designs.”

Our study establishes that even the small to medium-size firms that typify independent businesses are able to accommodate higher minimum wages with new revenues, leaving owners' profits unchanged. Workers receive larger paychecks and face flat employment rates in part due to worker reallocation from independent businesses to C-corporations. Higher worker retention and productivity allow incumbent owners to fully escape the economic burden of minimum wage increases. More broadly, minimum wages reshape the productivity distribution of highly exposed industries. Beyond curtailing entry, by changing the cost structure of industries that rely heavily on low-wage labor, minimum wage increases also affects the types of firms that enter with implications for incumbents and the industry overall.

II. Measuring Effects of the Minimum Wage with Tax Data

To estimate how independent businesses accommodate the increased labor costs associated with higher wage floors, we leverage administrative tax data to combine worker-level measures of earnings and employment with firm-level measures of revenues, costs, and profits.

II.A. Independent Businesses

To study independent (i.e., privately owned), predominantly small and medium-size businesses, we focus on firms organized as "pass-throughs" for tax purposes. Pass-throughs are privately owned businesses with legal forms including S-corporations, partnerships, and Limited Liability Companies (LLCs).[6] Pass-throughs are a large part of the U.S. economy. In 2015, they made up 78% of non–sole-proprietorship businesses and accounted for 46% of private-sector employment and 52% of business income. Pass-throughs are the predominant organizational form for small and medium-sized businesses in the United States. They

6. The name "pass-through" comes from the tax treatment. For these private firms, business income is not taxed at the entity level but is "passed through" to the tax returns of the firm owners. This contrasts with publicly traded C-corporations where business income is subject to the corporate income tax.

represent 79% of firms with fewer than 20 employees, 72% with 20–99 employees, 61% with 100–500 employees, and 77% of all firms with fewer than 500 employees.[7] Pass-throughs are smaller than publicly traded firms on average, but they represent the majority business form in all two-digit NAICS industries except utilities and management of companies and enterprises. Throughout the article, we simply refer to these firms as independent businesses or firms.[8]

We focus on independent businesses for two reasons. First, our linked firm-worker data are particularly well suited for cleanly estimating firm responses among these businesses. For large corporations, it is difficult to consistently link workers to their employers because of complex parent-subsidiary relationships and the use of third-party payroll services. Moreover, it is difficult to precisely link the operations of a firm (or the resulting revenues, costs, profits and employment patterns) to a specific location, which is essential for our research design. Second, the effect of minimum wage policies on independent businesses is understudied relative to their importance to the economy, despite often being at the center of debates about how minimum wages affect firms. The administrative tax data provide a unique opportunity to conduct a comprehensive analysis of how these businesses accommodate the cost shock associated with minimum wage increases.[9] Finally, we note that in the individual-level analyses, we sample all relevant workers regardless of whether they work at independent businesses or large corporations (Section V).

7. See Online Appendix Table C.4 for corresponding statistics. See Smith et al. (2019), Yagan (2015), Cooper et al. (2016), and Risch (2024) for further descriptions of pass-throughs and their owners relative to other business forms in the United States.

8. We refer to these businesses as "independent businesses" rather than "pass-throughs" because the salient feature for our purposes is that these are privately owned and operated firms, as opposed to large, publicly traded corporations with dispersed ownership and access to public capital markets. Pass-through is terminology related to the tax treatment of the business's income, which is not central to the discourse around the effect of minimum wages on businesses.

9. Industry data and data derived from financial statements generally cover large public corporations rather than small and medium-sized firms. Scanner data similarly come from larger retailers, and household surveys rarely have information on employer size.

II.B. Administrative Data: Firm-Worker Panel

Administrative tax data provide comprehensive information on all independent businesses and their workers during the study period. Firm and worker information is drawn from the universe of deidentified administrative tax data. We use a 100% sample of pass-through firms in "treatment" and "control" states (definitions in Section III) in each year from 2010 to 2019. For each firm-year, we collect information from the firm's annual income tax return, including revenues, cost of goods sold (COGS), various deducted operating expenses, net profits, and industry. We use these variables to construct performance measures like value-added (revenues less COGS) and materials costs per dollar revenue (COGS plus other deductions scaled by revenue). We link each firm to all workers that receive wage and salary payments in that year (through Form W-2). The worker data are used to define employment and total wages and salaries paid (total wage bills) at the firm level and to estimate within-firm changes in employment relationships (e.g., worker entry and separations, and distributions of earnings and ages).

The linked employer-employee data allow us to estimate various margins of response among affected firms and their workers, providing a comprehensive picture of how independent businesses respond to minimum wage increases. We construct firm panels to credibly estimate how firms respond when minimum wages are raised, and having the universe of independent businesses also provides a representative cross section of firms in each year, allowing us to estimate entrance and exit effects and how the characteristics of all independent businesses evolve over time. In our primary analyses, we focus on firms in highly exposed industries (Section II.C), which comprise 135,163 firms per year in the balanced panel and approximately 271,000 firms per year in the full sample. We use firms in non–highly exposed industries for placebo tests (approximately 1,386,000 firms per year).[10]

To complement our firm-level analyses, we construct two individual-level panels to estimate effects on potentially vulnerable workers. The first panel is a 2% random sample of all "low-earning" workers in treatment and control states in the year before the minimum wage increases. This data set spans

10. Further details of the data creation and sample are in Online Appendix L.

low-earning workers from all industries and firms, including large C-corporations that are not included in the firm analysis. This provides a more comprehensive analysis of individual earnings and employment effects, and contextualizes the firm-level effects observed among independent businesses. Second, we draw a 2% random sample of young individuals (ages 16–26) in treatment and control states in the year before the minimum wage increases, regardless of whether they are employed prior to the reform. We discuss these individual data in more detail in Section V, when we present the accompanying analyses.[11]

II.C. Identifying Industries Employing Minimum Wage Workers

Rather than being spread through the economy, minimum wage workers in the United States are concentrated in a handful of industries. We focus on these "highly exposed" industries to measure how minimum wage increases affect firms and the workers they employ in the industries where wage-floor policies are likely to meaningfully affect operations.

We use publicly available data to determine which industries employ enough minimum wage workers to be affected by policies to raise the minimum wage. We turn to public data because although the tax data afford detailed earnings histories for workers and comprehensive income statement descriptives for firms, they do not record hourly wages, making it challenging to precisely estimate exposure at the firm level. Specifically, we use the Current Population Survey Monthly Outgoing Rotation Groups (CPS MORGs; U.S. Census Bureau and BLS, 2012–2015), which describe the hourly wages of workers paid by the hour. Using the CPS MORGs data, we calculate the share of nonallocated hourly wage workers in each four-digit industry paid less than the prevailing minimum wage, pooling across all treatment states and cities in the year before each jurisdiction's policy change. As such, we effectively aggregate worker counts from different calendar years to appropriately measure industry shares just before the policy changes.

11. We also construct a repeated, randomly sampled, 2% cross section of teens in each state and each year to understand how teens entering the labor market after the minimum wage change might be affected by policy. The samples are detailed further in Online Appendix L.

We focus on four-digit industries employing at least 1% of minimum wage workers. These industries collectively account for more than two-thirds of minimum wage workers, with "restaurants and other food services" alone accounting for 42% of workers paid the prevailing minimum wage or less. Additional details, along with a full list of industries used in the analysis, are provided in Online Appendix B.

III. State Minimum Wage Changes and Empirical Strategy

III.A. Policy Details

Our analysis examines the effects of 19 minimum wage increases that began between 2013 and 2016 and did not exempt small firms, which includes increases among 17 states, Washington, DC, and the city of Chicago. We focus on minimum wage increases of at least $1 that began between 2013 and 2016 to ensure our 2010–2019 sample period spans enough years before and after the policy change to assess pre-trends and measure outcomes several years after the event.[12] Figure I shows the changes in minimum wages from these 19 policies over our sample period; Online Appendix Table A.1 shows the dollar and percentage changes in minimum wages from the pre-reform year to four years later, and Online Appendix Table A.2 provides additional policy details for each increase.[13]

For most reforms, the initial minimum wage increase was the first part of a larger, phased-in minimum wage hike. These phase-ins mirror federal proposals to raise the minimum wage over several years. The state increases were substantial. The average minimum wage increase was $2.59 or 34.1% ($2.64 or 34.6% firm weighted) four years after the initial increase. Across all treated states, in the year before the policy changes, ap-

12. This is a particularly clean period to estimate the effects of state minimum wage changes because it follows the last federal minimum wage increase in 2009 and offers a sufficient window after the state minimum wage changes and before the COVID pandemic began to affect labor markets in 2020.

13. We treat Maine's 2017 minimum wage increase as starting in 2016 because Portland, the state's largest metropolitan area, increased its minimum wage from $7.50 to $9.75 in 2016 in anticipation of the state-level change. The results are qualitatively and quantitatively robust to treating Maine's minimum wage increase as occurring in 2017 or dropping Maine from the sample. We exclude Arizona's 2017 $1.95 increase in its minimum wage as the Arizona wage floor does not apply to firms with revenues less than $500k

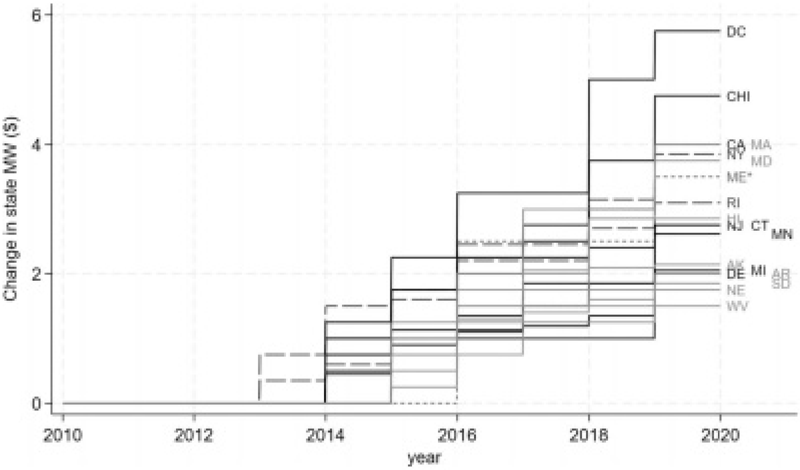

FIGURE I

State Minimum Wage Changes

This figure tracks changes in the statutory minimum wage in the 18 states and the city of Chicago, whose policy changes provide the identifying variation for the analysis. These states all increased their minimum wages starting between 2013 and 2016. Most minimum wage increases were phased in as indicated by the stepwise changes over time. *Maine's state minimum wage increased in 2017, but Portland, Maine's largest city, increased it's minimum wage in 2016 in anticipation of the state change, so we treat the pre-reform period for Maine as the years before 2016.

proximately 31.4% of hourly workers (or 18.46% of all workers) were paid wages below the new minimum wage.[14] Our event-study analysis traces the collective impact of these wage-floor changes over time with the estimates four years after the initial increase providing the most complete measure of the cumulative effect. Because firms may anticipate scheduled or future minimum wage increases, our estimates may attribute too strong a response to the actual change in minimum wages in any given year.

14. We use the CPS MORGs data to calculate the share of affected workers. To incorporate phase-ins, we use the minimum wage in time $s+4$ as the new minimum wage for each jurisdiction.

III.B. Identification

Using the policy variation described above, we define treatment and control groups of states based on whether they raised their minimum wages during our sample period. The treatment states are shown in Figure I, and the control group consists of 22 states that did not legislate a minimum wage increase between 2011 and 2019.[15] Our control group thus represents a set of "clean controls" that had no minimum wage changes during this period, which facilitates unbiased estimation in a stacked event study.[16]

To estimate the effect of the minimum wage increases, we use an event-study design to compare outcomes of firms and workers in treated and untreated states over time. This design helps assess the validity of comparing firms in treated and untreated states by estimating whether trends moved in parallel prior to the reforms. Our workhorse model is a panel stacked difference-in-differences (DD) specification of the form:

$$y_{jct} = \alpha + \sum_{s=-4, s\neq -1}^{4} (\beta_s treat_{jc} + \Gamma_s X_{jc}) \times year_{s=t} + \delta_{ct} + \psi_{jc} + \nu_{jct}, \tag{1}$$

where j indexes the firm, t the year, and c the "cohort" associated with the year of the initial minimum wage increase. We use a stacked design, meaning that for each treatment

15. The control states are Alabama, Georgia, Idaho, Illinois (excluding Chicago), Indiana, Iowa, Kansas, Kentucky, Louisiana, Mississippi, Nevada, New Hampshire, New Mexico, North Carolina, North Dakota, Oklahoma, Pennsylvania, South Carolina, Tennessee, Texas, Utah, and Virginia. Note that Illinois and Nevada had small minimum wage increases in the first year of our sample period, 2010, of \$0.25 and \$0.70, respectively. In our regression specifications, we include an indicator variable for these state-years, meaning that these control states do not contribute to the counterfactual for event-year $s-4$ for the 2013 cohort of minimum wage increases.

16. Recent econometric advances have shown that causal estimates from staggered DD designs can be biased if treatment effects are heterogeneous in units over time and previously treated units are used as controls for later-treated units, which can result in negative weights in OLS estimation (Callaway and Sant'Anna 2021; Goodman-Bacon 2021). Our estimation strategy obviates these concerns by using a set of never-treated units as controls for all treated units in each period, or "clean controls" (Cengiz et al. 2019).

cohort, there is a full set of control firms from all control states and estimates are in event time, s, relative to the cohort year.[17]

The DD estimator β_s represents the differential in the average outcome (y_{jct}) between firms in treated and untreated states relative to the pre-reform base year, $s-1$; $treat_{jc}$ is an indicator for the firm operating in a state with a minimum wage increase. The main specification includes firm-cohort fixed effects (ψ_{jc}) and cohort-by-year fixed effects (δ_{ct}). A vector of fixed firm and market characteristics (X_{jc}) is included to ensure comparisons among similar firms in similar markets, and it is flexibly interacted with year to allow for differential time trends among different types of firms. In the firm-level analyses, the controls included in X_{jc} are categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates, unless otherwise stated.[18] Standard errors are conservatively clustered at the state-by-cohort level.[19]

For our main analyses in Sections IV.B and IV.C, we estimate the effect on firm outcomes, z_{jct}, scaled by baseline revenues, that is, $y_{jct} = \frac{z_{jct}}{revenue_{jc,s-1}}$. These regressions are weighted by baseline log revenues. Occasionally, we estimate percent changes in a specific firm outcome, in which case we scale the contemporaneous value by the base year value, that is, $y_{jct} = \frac{z_{jct}}{z_{jc,s-1}}$, and regressions are weighted by the log of the baseline value of the corresponding variable. As such, all regressions are dollar or employee

17. That is, there is a set of firms from all control states for each of the 2013–2016 treatment cohorts. Each set of control firms has a unique event time relative to the associated cohort, spanning nine years around each cohort year. For example, the 2014 cohort treatment firms are those that first raise the minimum wages in 2014, which we call event time $s=0$, and the base year is the year prior to the minimum wage changes, 2013, or year $s-1$. For the associated control firms in this cohort, event time is assigned accordingly, 2014 is $s=0$ and 2013 is $s-1$. For the 2015 cohort, control firms are from the same set of states, but event time is around the 2015 minimum wage increase, so 2015 is $s=0$ and 2014 is $s-1$.

18. For analyses where we aggregate above the firm level (Figure VI; Table V, first row) or when using unbalanced panels in which firms enter the sample after the pre-reform period (Table III; Table V, second row), we exclude the baseline firm-level controls for firm-size and value added per worker.

19. We winsorize all raw reported firm values from above and below at the 1% level. Winsorization is standard when working with the population business tax returns. See, for example, Yagan (2015), DeBacker et al. (2019), and Kline et al. (2019).

TABLE I
SUMMARY STATISTICS: INDEPENDENT BUSINESSES IN HIGHLY EXPOSED INDUSTRIES

	Means (base year)		Medians (base year)	
	Treatment	Control	Treatment	Control
Revenue ($)	1,814,117	1,739,562	784,600	766,300
Wage bill ($)	340,283	312,326	115,800	113,300
Value added ($)	985,970	926,238	427,100	406,850
Owner income ($)	124,445	121,421	54,280	51,550
Employees	43.9	52.4	14	16
Young workers (16–26)	22	28	5	6
Share low-earning workers	0.28	0.31	0.26	0.30
Wage bill / revenues (dollar weighted)	0.19	0.18	0.17	0.18

Notes. This table reports summary statistics for the balanced estimation sample of 134,974 independent businesses in highly exposed industries in treatment and control states measured in the base year, defined as the year before the minimum wage increase for that state. For confidentiality, dollar values of medians are rounded to $100 or $10 and statistics on the number of workers include more than 10 firms with the reported value of workers.

weighted, presenting the policy-relevant estimates of the average effects.

When conducting individual-level analyses (Section V), we use the same regression framework, with slight modifications:

$$y_{ict} = \alpha + \sum_{s=-4, s\neq -1}^{4} (\beta_s treat_{ic} + \Gamma_s V_{ic}) \times year_{s=t} \tag{2}$$
$$+ \delta_{ct} + \rho_{ic} + \nu_{ict}.$$

First, for individual analyses we use a similar set of controls (V_{ic}), but drop the controls for firm size, industry, and value added, and instead include controls for individual age and age squared. Second, we include an individual-cohort fixed effect (ρ_{ic}) rather than a firm fixed effect. In addition, we use a number of linear probability models (LPMs) where the outcome is an indicator variable for being employed or for being employed in a certain type of firm, in which case the coefficients are interpreted as percentage point changes for the treatment group relative to the control group.

Table I reports key characteristics of the sampled firms with means presented in the left panel and medians in the right. The summary statistics show that the firms in treatment and control states are quite similar in terms of observable characteristics before any regression adjustments. The firms are relatively small, with average revenues of about $1.75 million, net income of about $120,000, and 40–50 workers per year on average. The medians are below the means, highlighting the predominance of small firms in these industries, but with a tail of larger independent businesses.

III.C. Implications of the Research Design for Understanding Minimum Wage Responses

Our identification strategy compares firms and individuals in states that raised their minimum wages to similar firms and individuals in states that left their wage floors unchanged. Because our estimation strategy does not rely on exploiting variation by baseline firm-level exposure, we are able to separately estimate the effects of increased minimum wages on firms that were active in the pre-period and how the minimum wage shapes aggregate outcomes and firm characteristics in exposed markets through firm-entry and exit dynamics.

The principal threat to identification is policy endogeneity.[20] One form of endogeneity arises if states that elect to raise minimum wages at a given time are on different paths than those that do not, for example, if minimum wages are increased in states with faster growing economies. This would be a violation of the standard parallel-trends assumption for DD designs and is a legitimate concern. We use pre-trend analyses for all of the primary outcomes to assess the plausibility of this assumption in our setting. The second identifying assumption is that there are no other concurrent changes in the economy that would differentially affect outcomes in treated and control states. This is a general challenge for designs with state-level variation. We provide evidence that our estimates are consistent with responses to minimum wage policies, for example, by examining the distribution of wage changes and types of workers that are affected by firm-level employment adjustments, and by showing that differential outcomes between treatment and control states are concentrated only in industries highly exposed to minimum wage changes.

20. Research centered on minimum wage increases during recessionary periods finds somewhat larger employment impacts (Clemens and Wither 2019) and effects on firm finances and exit rates (Chava, Oettl, and Singh 2019).

IV. Firms' Adjustments to Higher Minimum Wages

Our firm-level analysis provides granular insight into how independent businesses adjust to higher wage floors and the ultimate effect on highly exposed industries. First, we use a balanced panel to learn how existing firms accommodate the new labor cost shock. The balanced panel allows us to focus on within-firm behavioral responses, separate from compositional shifts driven by entry and exit. We examine how firms adjust their labor inputs, estimating changes in the number of employment relationships and the types of workers and jobs that are affected, and we analyze how firms finance the new labor costs to understand who bears the burden of the wage increases (Sections IV.A–IV.C). Then we use the full unbalanced panel of all active firms in exposed industries in all years to assess how minimum wages affect the viability of independent firms and how changes in firm composition shape aggregate outcomes in highly exposed industries (Sections IV.E and IV.F).

IV.A. Minimum Wages and Firm Employment

Our firm-worker panel tracks employment relationships between firms and all workers who receive wage and salary compensation from the firm in a given year, including part-time or part-year employees.[21] The first panel of Figure II traces the average change in the net number of employment relationships, as well as the associated changes in new (entrants) and existing employment relationships (separations), as estimated using equation (1). The average independent business subject to a higher wage floor does not lay off workers it employed before the minimum wage increase. Firms in treated states, however, reduce hiring. By four years after the minimum wage increases, firms in highly exposed industries hire, and thus have employment relationships with, roughly 1.5 fewer workers a year than similar firms in control states on average. This corresponds to an own-wage employment elasticity of -0.245 (std. err. $= 0.134$), which is small in magnitude and in line with previous studies that do not focus on the

21. This is typically the case with administrative data that describes only employment and compensation rather than hours worked by employees over the reporting period (here one year).

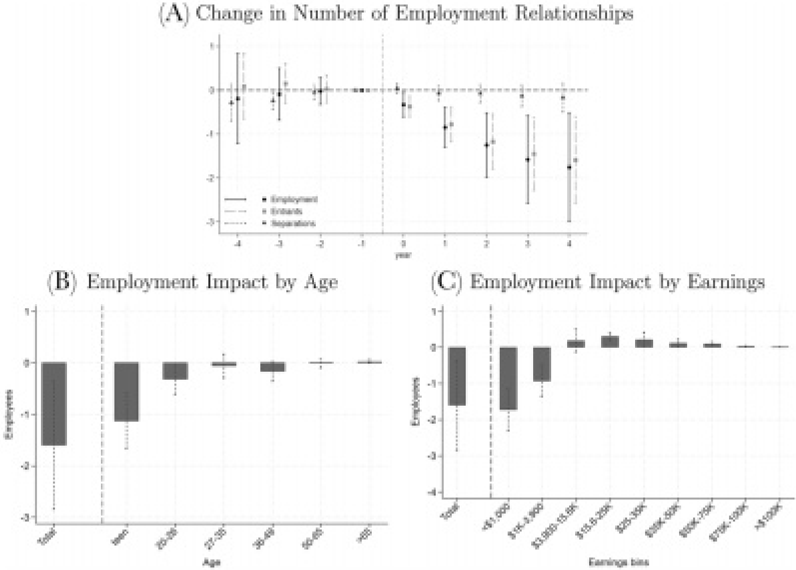

FIGURE II

The Change in Employment Relationships Due to Minimum Wage Increases

This figure describes the change in the number and composition of employment relationships due to the minimum wage. Our employment measure counts the number of W-2s attached to a firm each year. Time $s = -1$ is the year before the minimum wage increase for each cohort of states. Panel A traces how minimum wage increases affect the number of employment relationships of the average firm as well as the number of new (entrants) and departing (separations) workers. Plotted coefficients are estimates of β_s from equation (1), which convey the change in employment relationships relative to base year $s - 1$, for the average treated firm compared to the average control firm. Entrants capture the number of new hires, defined as employees of the firm in year s that were not with the firm in year $s - 1$, while separations describe those at the firm in year $s - 1$ that were not with the firm in year s. This decomposition is such that the coefficients from the entrants and separations specifications sum to the net employment response. Panels B and C decompose the $s - 1$ to $s + 4$ change in total employment relationships by worker age and annual earnings, respectively. Each bar is the difference-in-differences estimate from a regression where the outcome is the five-year change in the number of workers in each category relative to the base year number of workers for firm j. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Regressions are weighted by base year firm employment. Employment counts are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

small and medium-sized firms that typify independent businesses (see Online Appendix Figure D.1).[22]

Reduced hiring is highly concentrated among teenagers and low-earning jobs. Figure II, Panels B and C decompose the change in employment relationships by employee age and earnings. Firms subject to higher minimum wages primarily hire fewer workers under 20 years of age. The missing hires consist entirely of workers who would have been paid less than \$3,900 annually—with the majority (67%) earning less than \$1,000 a year. In short, higher minimum wages lead these firms to reduce very part-time, less experienced labor inputs.[23]

IV.B. Financing Higher Pay to Low-Wage Workers with Added Revenue

Because firms make only minor employment adjustments, we expect wage bills to rise sharply after the minimum wage increase. Figure III, Panel A plots coefficients from a firm-level event-study regression of equation (1) where the dependent variable is the ratio of the firm's wage bill to its base year revenue. The annual estimates measure the average scaled wage bill increase among firms in highly exposed industries operating in states that raised their minimum wages compared with similar firms in control states.

The estimates before the reform year ($s = 0$) show that firms in treated and untreated states had similar wage bill trends in the years before the minimum wage increases. Average wage bills then rise sharply among treated firms. Wage bills grow each year as minimum wage increases phase in; four years down the line,

22. We estimate firm-level own-wage elasticities by separately estimating the change in log employment (2.06%, std. err. = 1.03%) and the change in log average wages (8.11%, std. err. = 1.34%) using equation (1), then dividing the estimated change in log employment by the change in log average wages. Standard errors are calculated using the delta method. These estimates can be found in Online Appendix Table H.17 and Online Appendix Figure H.5. It is worth noting that these changes in the number of W-2s issued by highly exposed independent businesses may overstate the true change in the number of jobs at these firms, as minimum wage increases affect retention and worker churn. We consider these dimensions of minimum wage effects in Section V.D.

23. These results align with Wursten and Reich (2023), who find that the minimum wage's only disemployment effects are among teenagers at small businesses, and Dube, Naidu, and Reich (2007), who study restaurants and find no evidence of employment losses.

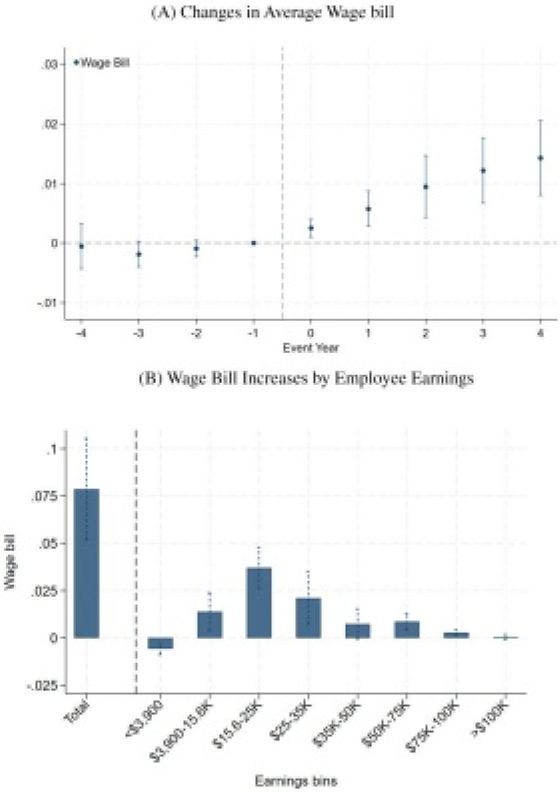

FIGURE III

The Impact of Minimum Wage Increases on Firms' Wage Bills

This figure presents the estimated effects of the state minimum wage increases on average firm wage bills overall (Panel A) and by worker earnings (Panel B) for independent businesses in highly exposed industries. Panel A traces out the difference over time in average scaled wage bills relative to the base year $s-1$ between firms in treated and untreated states. Estimates are from a regression of equation (1) where the dependent variable is the firm wage bill scaled by baseline revenue in year $s-1$ and plotted coefficients are estimates of β_s. Panel B shows the decomposition of the total five-year wage bill response from year $s-1$ to year $s+4$ across the worker earnings distribution. The total wage bill effect is shown on the left, and the gains by earnings range are on the right. Each bar is the difference-in-differences coefficient estimate from a regression where the outcome is the five-year change in the wage payments in each earnings bin scaled by baseline revenue for firm j. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and wage bill–to–base year revenue ratios are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

the average treated firm pays 1.43% more in wages as a share of baseline revenues.[24]

Raising the minimum wage directly increases the hourly pay of workers for whom the floor binds but may also indirectly affect the incomes of other workers. Firms may partially offset the added costs of higher minimum wages by reducing the compensation of higher-earning workers. Alternatively, increased pay to the lowest earners may boost pay higher up the distribution as firms maintain wage differentials by occupation or seniority.[25] Figure III, Panel B examines how higher minimum wages affect the distribution of compensation across workers in a firm. Annual compensation rises substantially for workers earning between \$3,900 and \$35,000 a year, with the largest gains accruing to workers earning the equivalent of working full-time at the minimum wage. There is no evidence of reduced earnings for those higher up in the earnings distribution, ruling out redistribution from middle- or high-income employees. Instead, earnings also rise for workers earning slightly above full-time at the new minimum wage, suggesting potential spillovers to low-earning workers above the statutory wage floor.[26]

Since independent firms do not substantively shed workers or reduce compensation to higher-income workers in re-

24. The observed wage bill change subsumes any changes to their input mix that firms make in response to the higher wage floor. Because we do not observe large employment changes, these behavioral responses may be less substantial. They will, however, include any other adjustments that firms make to the hours employees work and the type and effort of work they do. To benchmark the estimated effect size, Online Appendix Table H.17 shows that this corresponds to a 6% wage bill increase by year $s + 4$. On average, approximately \$54,000 of the wage bills of treatment firms are from earnings paid to low-earning workers at baseline. On average minimum wages increase by 34.6% by year $s + 4$ (Online Appendix Table A.1). A 34.6% increase in low-earning worker earnings is \$54,000 $\times$ 0.346 = \$18,684, which is a 5.5% increase in the total average baseline wage bill from Table I, quite similar to the estimated 6% increase.

25. In the spirit of Clemens, Kahn, and Meer (2018), we use measures of compensation outside of wages and salary to learn if minimum wage increases reduce noncash compensation. These estimates, reported in Online Appendix Table E.6, show that higher minimum wages have no discernible effect on firm deductions for other worker benefits, which include health insurance and other in-kind benefits, but very slightly raise pension contributions.

26. This pattern also serves as a validity test for the results, showing that the wage bill increases are driven by the low-earning workers most likely to be affected by minimum wage policies, and not by high-earning workers where changes may be more likely to be driven by other policies or market shocks.

sponse to minimum wage increases, they must offset higher wage bills in other ways. Prior work suggests that firms are often able to pass the cost of minimum wage increases on to consumers through higher prices (Sorkin 2015; Harasztosi and Lindner 2019; Link 2019; Ashenfelter and Jurajda 2021; Leung 2021; Renkin, Montialoux, and Siegenthaler 2022) although there is also evidence of reduced profits (Draca, Machin, and Reenen 2011; Ganapati and Weaver 2017; Harasztosi and Lindner 2019).[27] Although tax data do not allow us to separate quantities and prices, we can examine how revenues evolve to offset higher labor costs after the minimum wage increases.

Figure IV plots the average annual change in revenue relative to baseline among firms subject to higher minimum wages compared with firms in control states. After following a pretrend similar to that of untreated firms, revenues among treated firms rise starting the year after the minimum wage increase. Four years later, the revenue of the average firm subject to a higher wage floor had grown roughly 3.31% more than the revenue of firms in states that did not raise their minimum wages.

These added revenues shape the ultimate impact of minimum wage increases on owner profits. Figure V plots the evolution of owner profits as a share of baseline revenues among firms subject to higher minimum wages relative to firms that were not. Owner profits capture the total return to owners of the business, combining business profit with salary payments to owner-operators. Four years after the minimum wage increases, average owner profit is unchanged, with a point estimate of 0.0002 (std. err. = 0.0029), ruling out losses larger than 0.37% of baseline revenues with 95% confidence.

IV.C. Understanding the Economic Burden of Minimum Wage Increases

Beyond measuring how wage bills, revenues, and owner profits evolve after the minimum wage increase, we combine these es-

27. Studies using scanner data such as Leung (2021) and Renkin, Montialoux, and Siegenthaler (2022) naturally focus on products made and retailed by the large firms that typically report their data to Nielsen, Kantar, or other scanner data aggregators. As such, these estimates may be less applicable to small and medium-size firms or service-sector firms that account for the majority of minimum wage workers.

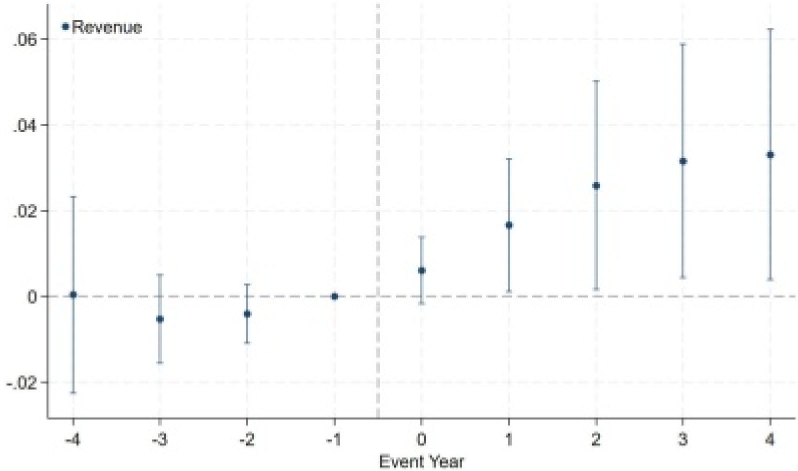

FIGURE IV

The Impact of Minimum Wage Increases on Firm Revenues

This figure plots the estimated effects of the state minimum wage increases on firm revenue. Plotted coefficients are estimates of β_s from a regression of equation (1) where the outcome is firm revenue scaled by revenue in base year $s-1$. The coefficients trace the difference over time in revenue relative to the base year $s-1$ between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and scaled revenue ratios are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

timates with changes in other tax-reported expenditures to trace the incidence of minimum wage increases. Table II presents estimates of the four-year impact of minimum wages on different income statement items: the components of variable costs, revenues, and owner profits, all scaled by baseline revenue to facilitate comparisons. For wage bills, revenues, and owner profits, these estimates match the $s+4$ estimates from Figures III to V.

In addition to wage bills, our data track expenditures on nonlabor inputs, such as the COGS, which capture intermediate goods, and "other deductions," which capture spending on

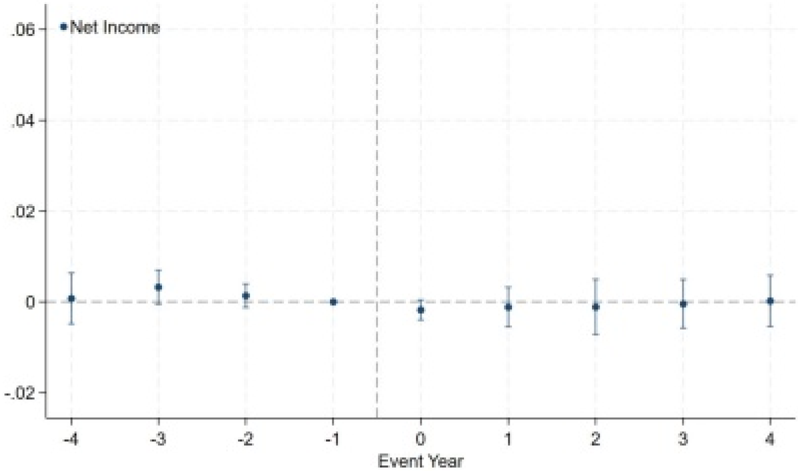

FIGURE V

Impact of Minimum Wage Increases on Firm Profit

This figure plots the estimated effects of the state minimum wage increases on firm profits (net business income) scaled by base year revenue. Plotted coefficients are estimates of β_s from a regression of equation (1), which trace out the difference over time in average scaled profit relative to the year before the policy change between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and profit-to-base year revenue ratios are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

other nonlabor inputs.[28] Minimum wage increases can affect spending on nonlabor inputs if firms substitute away from labor or face higher prices from upstream firms. Higher spending on nonlabor inputs may also reflect productivity gains from minimum wage increases, as documented by Coviello, Deserranno, and Persico (2022) and Emanuel and Harrington

28. Other deductions include "total allowable trade or business deductions that aren't deductible elsewhere." Eligible deductions include supplies used and consumed in the business, certain business start-up and organizational costs, insurance premiums, legal and professional fees, and utilities. We also estimate the impact of minimum wage increases on investment and find no substantive effect as reported in the depreciation line item of Online Appendix Table E.6, which includes immediately expensed investment (Section 179).

TABLE II
THE INCIDENCE OF MINIMUM WAGE INCREASES

	All exposed	Restaurants	Other/retail
Cost burden to firm			
Wage bill	0.0143***	0.0201***	0.0076***
Financed by consumers			
Revenue	0.0331**	0.0294**	0.0359**
COGS	0.0131***	0.0053	0.0210***
Other deductions	0.0035	0.0029	0.0039
Financed by owners			
Owner profits	0.0002	−0.0001	0.0003
Net share by consumers	100%	100%	100%
Net share by owners	0%	0%	0%

Notes. This table compares estimated changes in wage bills and other input costs to estimated revenue increases to understand who bears the burden of higher wage floors. Net shares of the burden of the higher labor costs borne by consumers and owners are calculated using the regression estimates. Each estimate is from a firm-level regression using the balanced panel of equation (1) where the outcome is the variable defined in each row scaled by baseline ($s-1$) revenue. Because we use the balanced panel, these estimated effects are not affected by extensive-margin responses. The coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. There are 134,974 firms in the *All exposed* sample, 68,193 of which are restaurants. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (no. of workers), deciles of baseline firm value added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. All raw reported firm values are winsorized from above and below at the 1% level. Regressions are weighted by log baseline revenue. *** $p < .01$, ** $p < .05$, * $p < .1$.

(2025), with spending on COGS and equipment rising with added sales.

The average independent business operating in a highly exposed industry not only spent more on wages after the minimum wage increase but also increased its expenditures on COGS by 1.31% of baseline revenue. Higher spending on nonlabor inputs is differentially concentrated across industries. Labor is a relatively more important input among restaurants, which collectively employ 42% of all minimum wage workers, whereas intermediate materials costs (e.g., inventories) are a larger share of variable costs for highly exposed industries outside of restaurants, largely retail. For restaurants, wage payments are 25% and COGS are 57% of revenues at baseline; for other industries, wages are only 13% of revenues and COGS are 65% (Online Appendix Table G.14); for retail, wage payments are only 9% of revenues and COGS 77% (Online Appendix Table G.16). As such, minimum wage increases raise restaurant wage bills substantially as a share of baseline revenue—by 2.01% four years after the wage-floor increase—while COGS rises by only 0.5% of baseline revenues. In contrast, among highly exposed industries outside of restaurants, COGS rise by 2.1% of baseline revenue, and labor costs increase by 0.76% of baseline revenue.[29]

Regardless of industry, higher minimum wages increase variable costs by similar amounts, roughly 2.8% to 3.1% of baseline revenue. As Table II shows, this increase in costs is fully financed with new revenues such that owners bear no economic burden. For a treated company, per dollar of baseline revenue, labor costs rose by 1.43 cents and nonlabor costs rose by 1.66 cents; revenue, however, increased 3.31 cents, leaving profits wholly unchanged. In short, even for the small- and medium-sized firms that typify independent businesses, consumers fully finance the higher pay to workers generated by minimum wage increases.[30]

IV.D. Robustness of Firm Analysis

In the Online Appendix, we examine the robustness of the firm estimates that underlie the preceding analyses. First, we show that the results are not overly sensitive to any specific controls used in the firm-level regressions; the controls correct for some selection in pre-trends but do not drive the qualitative results (Online Appendix Figure H.4). Online Appendix Table E.7 shows that the firm results are very similar when using an unbalanced panel that allows firms to exit, which is unsurprising given the lack of differential exit rates documented in the following subsection. Online Appendix Table H.17 and Online Appendix Figures H.6 and H.7 show that the results and pretrends are consistent when estimating percent changes rather than outcomes scaled by revenue. We examine the heterogeneity of our estimates by baseline size (Online Appendix Table F.8) and show the results—modest employment adjustments and

29. Online Appendix Table H.17 reports effects on firm outcomes in percent changes. While wage bills only rise modestly as a share of baseline revenues among non-restaurants, this corresponds to a 4.4% increase in wage bills on average. The modest change relative to revenues reflects the low baseline share of wage bills to revenues. For restaurants, wage bills rise 6.61%. Online Appendix Tables G.15 and G.16 provide further detail by industry.

30. Independent firms hire fewer part-time workers after minimum wage increases; whether this group bears a portion of the costs depends on whether affected workers find employment elsewhere and is examined in detail in Section V.

revenue increases sufficient to fully cover wage and other cost increases—are consistent across the firm size distribution, including for the smallest independent firms. Although the magnitudes of revenue and cost effects differ somewhat, revenues rise sufficiently to fully offset cost increases across the baseline productivity distribution (Online Appendix Table F.10) and among firms with different baseline reliance on low-earning labor (Online Appendix Table F.9).[31] Roth et al. (2023) show that even with parallel trends conditional on covariates X_j, linear regressions can yield inconsistent estimates if treatment effects are heterogeneous in X_j. Online Appendix Table J.24 shows that the results are very similar using "regression-adjusted" estimators of the treatment effects, which are consistent under weaker homogeneity assumptions.[32]

As a further robustness test, we present results from a design using border counties following Card and Krueger (2000), Dube, Lester, and Reich (2010, 2016) and Allegretto et al. (2017) (Online Appendix M, Online Appendix Table M.28). Although the magnitudes of the estimated effects on firm outcomes are somewhat attenuated, the qualitative results are very similar when using the border firms as when using the full sample. Revenues rise enough to offset the higher wage and nonlabor costs such that owners' profits do not decrease. The estimated employment effects and corresponding own-wage elasticities are also similar to those estimated using the full sample.

Finally, as a placebo test, we estimate the effects of the minimum wage increases on independent firms in industries largely unexposed to these policies (Online Appendix Figure H.8). We find no differential trends in firm outcomes before or after the minimum wage changes. There is also no effect on the average number of employment relationships among firms in these industries, with a point estimate of -0.191 (std. err. $= 0.152$) as of year $s + 4$. Taken together, these results support the claim that observed effects in highly exposed industries are driven by minimum wage

31. Own-wage elasticities are somewhat higher among larger firms and less productive firms.

32. Our use of nonparametric controls also mitigates this concern, as does the robustness across subsamples and when including various controls. Details of the "regression-adjusted" estimation are in Online Appendix J.

policies rather than differential concurrent economic changes in treatment and control states.

IV.E. Extensive-Margin Firm Responses

Although minimum wage increases have minimal employment effects at incumbent independent firms in highly exposed industries, and firm owners fully escape the economic burden of higher labor costs, we may worry that slimmer margins and unchanged fixed costs may lead some firms to shutter or forgo entry. To measure changes in the number of independent firms operating in treatment and control states, we create a collapsed data set that allows us to estimate differential entry and exit rates while controlling for relevant market characteristics. Concretely, we collapse the firm-level data set by cohort, year, treatment status, industry, and market characteristics. The collapsed data contain counts of active firms in each year that did and did not operate in the base year $s-1$, summing to the total number of active firms in each cell in each year. Using these data, we estimate the effect of the minimum wage on the number of active independent firms. We decompose the effect into the portion attributable to firm entry and exit using regressions in the style of equation (1), where the outcome variables are the numbers of (i) active firms in year t (all firms), (ii) active firms in year t that did not exist in the pre-reform year (entrants), and (iii) active firms in year t that existed in the pre-reform year (incumbents), each scaled by the total pre-reform number of active firms.[33]

Figure VI plots annual changes in exit, entry, and the total number of firms relative to base year, $s-1$. Minimum wage increases reduce the number of active independent firms in exposed industries by 1.97%. The decomposition shows that this is fully attributable to reduced entrance rates; firm exit is flat after the minimum wage increase. Our reasonably precise zero estimate rules out exit increases larger than 0.53% with 95% confidence. Ultimately, existing owners are able to finance higher pay to low-wage workers with new revenues, avoiding both profit decreases and business closures.

Exit may be the less sensitive extensive response margin in the medium run. Firms often sign long-dated leases, ren-

33. Online Appendix L.5 provides further details of the analysis using the collapsed data.

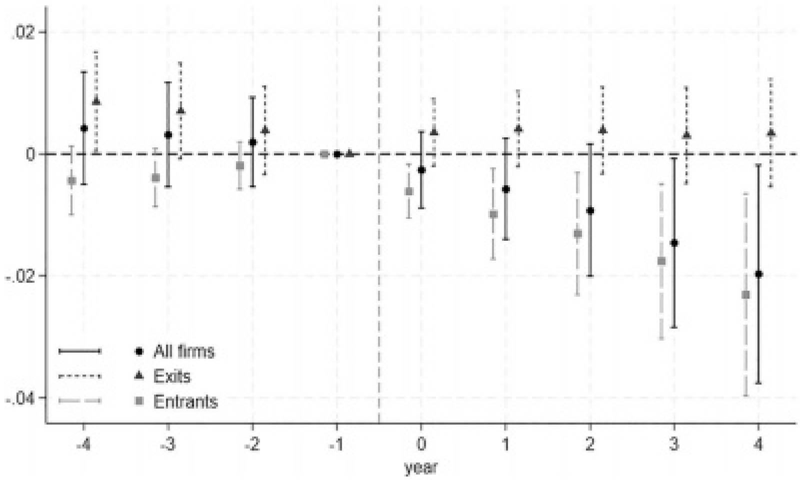

FIGURE VI

The Impact of Minimum Wage Increases on Firm Entry and Exit

This figure plots the impact of minimum wage increases on firm exit, entry and firm counts. Section IV.E and Online Appendix L.5 describe the collapsed data used for this analysis and the regression specification. Plotted coefficients are estimates of β_s and trace out the difference over time in the number of firms, entrants, and exits relative to the year before the policy change between treated and untreated states. In addition to cohort-by-year fixed effects, the specifications include baseline two-digit industry, quintiles of county density, quintiles of county employment rates, and quintiles of commuting zone pre-period firm churn—all flexibly interacted with year to allow for differential time trends. Regressions are weighted by the base year number of firms in each collapsed cell and standard errors are clustered at the state-by-cohort level.

dering their largest fixed costs sunk and leaving them willing to operate as long as sales cover their operating expenses. However, potential entrants may balk at the higher costs of operating in low-wage industries and opt not to enter. Although raising the minimum wage has no effect on exit, it meaningfully reduces entry. Four years after the minimum wage is raised, entry into industries highly exposed to wage-floor policy is 5.5% lower than in control states.[34] Slower entry meaning-

34. Four years after the minimum wage increases, 36.4% of active independent firms in treatment states are entrants, so the 1.97% reduction in total active firms is driven by a 5.5% reduction in entrant firms. Over this period, for independent firms in highly exposed industries approximately 8.6% of active firms in a given year were not active the previous year.

TABLE III

THE EFFECTS ON AVERAGE PRODUCTIVITY AND COST STRUCTURE

	Wage bill/ revenue	Material costs/ revenue	Value added/ worker	Profits/ revenue	Employees/ firm
All firms	0.0072***	−0.0048***	0.0462**	0.0031	−1.899***
	(0.00167)	(0.00150)	(0.0102)	(0.00454)	(0.0543)
% change	4.0	−0.8	4.6	3.7	−4.0
Entrants	0.0117***	−0.0157*	0.1550***	0.0070	−4.189***
	(0.00297)	(0.00847)	(0.0473)	(0.0111)	(0.969)
% change	6.1	−2.5	15.5	7.3	−9.8
Incumbents	0.0091***	−0.0033	0.0313**	−0.0064**	−1.496**
	(0.00163)	(0.00215)	(0.0149)	(0.00251)	(0.602)
% change	4.9	−0.5	3.1	−7.7	−2.1

Notes. This table shows estimated average changes in various metrics, estimated separately for all independent businesses in highly exposed industries, *Entrants* (firms that are active four years after the minimum wage increases (year $s+4$) but did not exist prior (year $s-1$)), and *Incumbents* (firms that were active in base year $s-1$ and remain active in year $s+4$). The full unbalanced panel includes ≈ 271,307 firms/year, the incumbent sample is 134,974 firms/year with entrants the remainder, 136,333. Each estimate is from a firm-level regression using an unbalanced panel of all active independent firms in highly exposed industries using equation (1). The coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. Each column represents an outcome variable. For value added per worker we use the natural log of the outcome to reduce noise and facilitate interpretation. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. Outcomes are winsorized from above and below at the 1% level. To facilitate interpretation we include implied percent changes. For *All firms* and *Incumbents*, these are the coefficients evaluated at baseline averages; for *Incumbents*, the percent change in employees is from the balanced panel regression estimate (Online Appendix Table H.17); for *Entrants* coefficients are evaluated at the control mean in year $s+4$. Standard errors are clustered at the state-by-cohort level. *** $p < .01$, ** $p < .05$, * $p < .1$.

fully affects these industries, ultimately reducing the number of firms.[35]

IV.F. Selection and Productivity

Beyond curtailing entry, by changing the cost structure of industries that heavily rely on low-wage labor, minimum wage increases also change the types of firms that enter with implications for existing firms and the industry as a whole. Tables III and IV examine how the distributions of performance measures change among firms in treated states relative to similar firms in states that did not adjust their wage floors.

35. For exit rates, which can be estimated using firm-level regressions, we validate the results using equation (1) with the standard set of firm and market controls and the outcome variable being an indicator for the firm operating in year t. Online Appendix Figure H.10 shows the results of this exercise, which are very similar to the main results using the collapsed data.

TABLE IV

COST STRUCTURE AND PRODUCTIVITY IMPACTS OF MINIMUM WAGE INCREASES

	Wage bill/revenue		Material costs/revenue		Value added/worker	
	Low (Q1)	High (Q4)	Low (Q1)	High (Q4)	Low (Q1)	High (Q4)
Entrants	−0.0236***	0.0271***	0.0324	−0.0267**	−0.0581***	0.0411***
	(0.00781)	(0.00775)	(0.0217)	(0.0124)	(0.0176)	(0.0128)
Incumbents	0.0021	0.0310***	0.0194***	0.0020	−0.0123***	0.0173***
	(0.00322)	(0.00502)	(0.00569)	(0.00329)	(0.00360)	(0.00718)
All firms	0.0015	0.0237***	0.0179***	−0.0011	−0.0215***	0.0122***
	(0.00288)	(0.00434)	(0.00585)	(0.00237)	(0.00327)	(0.00465)

Notes. This table describes changes in the distribution of labor shares, materials cost efficiency, and productivity among incumbent firms and entrants in treatment states relative to control states after the minimum wage increases. The full unbalanced panel includes ≈ 271,307 firms/year, the incumbent sample is 134,974 firms/year with entrants the remainder, 136,333. Materials costs per dollar revenue are the sum of COGS and other deductions scaled by revenue. Value added per worker is revenue less COGS scaled by the number of employees. Using the baseline distributions of wage bill per revenue dollar, materials costs per revenue dollar, and value added per worker, we define the top and bottom quartiles of labor use, other input efficiency, and productivity. We then estimate regression equation (1) where the outcome variable is an indicator for the firm being in quartile q of a specific measure in year $s + 4$. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include indicators for two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. *** $p < .01$, ** $p < .05$, * $p < .1$.

First, we estimate average changes in productivity and cost-structure measures (Table III) using an unbalanced panel of all active independent businesses in highly exposed industries. We report estimated changes four years after the minimum wage increases (year $s + 4$) for all firms, and separately for "entrants" (firms that did not exist prior to the minimum wage increase but were active in year $s + 4$), and "incumbents" (firms that were active pre-reform and remained active four years out). Among all active firms, we see the average cost-structure changes—wage bills rise while nonwage material costs decrease as shares of revenue. This pattern holds for incumbent firms, but the shift is stronger among new entrants. Average productivity, measured as value added per worker, increases significantly, again with particularly strong gains among entrants. Profit shares remain stable on average but decline among incumbents, consistent with our earlier finding that among these firms revenues rise to cover higher costs, while profits remain flat (Section IV.B). Finally, while reductions in employment relationships among all firms are similar to those observed for incumbents (Section IV.A), they are relatively larger among entrants, who account for a smaller share of firms (36%) and are smaller firms on average.[36]

To further examine how minimum wage increases reshape cost structures and productivity distributions, we use baseline quartiles of key performance measures like the wage bill–to–revenue ratio or value added per worker to define the top and bottom quartiles of labor use, other input efficiency, and productivity. We estimate the probability that a firm's performance four years after the minimum wage increase would rank in the top or bottom quartile of the baseline distribution.[37] These differential probabilities show how the distributions of firm characteristics evolve as a result of the minimum wage increases. Table IV reports estimates for entrants, incumbents, and highly exposed industries overall.

The distributions of costs and productivity shift after the minimum wage increases, driven partly by strong selection among entrant firms. Both entrant and incumbent firms are more likely to have high wage bill–to–revenue ratios after the policy change. New firms, in particular, are also less likely to rank among the lowest wage bill–to–revenue ratio firms. This evidence suggests that higher labor costs do not necessarily lead entrants to design their facilities to rely less on labor; instead, the firms that enter are precisely those that can cover the higher labor costs of a post–minimum wage increase market.

Firms offset the higher labor costs of the minimum wage increase in part by becoming more efficient. Measuring material costs as COGS plus other deductions, we find that both incumbent firms and entrants are leaner after the minimum wage increase. Incumbent firms are 1.94% more likely to be low cost, and entrants are 2.67% less likely to rank among the least cost-efficient firms. Firms finance higher minimum wages by generating more revenue without proportionately increasing input costs.

In addition to making firms leaner, minimum wage increases raise labor productivity, measured as value added per worker,

36. Online Appendix Table H.18 shows similar results when aggregating by industry, with and without market controls.

37. Concretely, we estimate regression equation (1) where the outcome variable is an indicator for the firm being in quartile q of a specific measure in year t.

TABLE V

Aggregate Effects on Independent Businesses in Highly Exposed Industries

	Revenue	Wage bill	Material costs	Profits	Value added/ worker
Aggregates (share of baseline revenue)	0.0263	0.0112*	0.0133	−0.0012	0.0805*
	(0.0294)	(0.0061)	(0.0168)	(0.0025)	(0.0467)
Averages (logs)	0.0432***	0.0767***	0.0266***	0.0278***	0.0464***
	(0.0048)	(0.0054)	(0.0059)	(0.0083)	(0.0044)

Notes. This table presents estimates from the full, unbalanced panel of independent businesses in highly exposed industries ($\approx$ 271,307 firms/year). *Material costs* are the sum of COGS and other deductions. *Value added/worker* is revenue less COGS scaled by the number of employees. The first row presents aggregate statistics summing over all independent firms in highly exposed industries. These estimates use collapsed state-by-industry data with state-by-industry totals of income statement items scaled by base year ($s-1$) total revenue as the outcome variables. Online Appendix L details the specification. All aggregate regressions are weighted by baseline total revenue. The lower row reports average log impacts in the unbalanced panel. The estimates are from a specification similar to equation (1), which includes cohort-by-year fixed effects and indicators for two-digit industry, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. All standard errors are clustered at the state-by-cohort level. *** $p < .01$, ** $p < .05$, * $p < .1$.

making it easier for firms to finance higher pay for workers. Both existing and new firms are more likely to fall into the high value added per worker group after the minimum wage increase. Entrants, in particular, are sharply selected; the average entering firm is 4.11% more likely to have high productivity and 5.81% less likely to have low productivity. Active firms, whether incumbents or entrants, are precisely those generating enough revenue per worker to cover the new higher labor costs.

Reduced entry and strong selection among entrants reshape how revenues are distributed in these highly exposed industries. Using a state-by-industry collapsed data set, we estimate how minimum wages affect aggregate measures across all independent firms in highly exposed industries. The upper row of Table V reports the effect of the minimum wage on aggregate outcomes scaled by baseline revenues. Many of the estimates are not statistically significant as they compare state aggregates and thus draw on relatively few observations; nonetheless, they provide suggestive evidence on the aggregate effects of minimum wage increases. Total revenues rose by 2.63% four years after the minimum wage increases, indicating that reduced entry did not fully offset higher revenues among operating firms. In aggregate, wage bills rise as a share of collective revenue, indicating that raising the minimum wage ultimately results in workers receiving a larger slice of a larger revenue pie. Material costs increase along with wage bills, but overall productivity improves enough to keep aggregate profits unchanged. These estimates confirm that in aggregate, total payments to workers rise and suggest that the burden is borne by consumers rather than owners.[38]

The lower row of Table V reports estimates of average log impacts using a firm-level unbalanced panel that includes all active independent firms in highly exposed industries in all years. These estimates subsume changes in firm composition due to differential entry. Estimates using the full population of firms confirm the findings from both the balanced panel and the collapsed aggregates. Average revenues and wage bills rise substantially, by roughly 4.3% and 7.7% respectively, and value added per worker rises enough on average to fully protect firm profits.[39]

The ability of firms to finance higher labor costs with increased revenues could reflect price pass-through, reallocation of demand toward (more productive) surviving businesses, or a combination of the two. Our results suggest that both mechanisms could matter. That revenues rise among incumbents and in the aggregate suggests that price pass-through plays a role, while the reduction in firm entry (and strong selection among entrants) point to demand reallocation. Because our data do not separately report prices and quantities, we cannot empirically distinguish these channels. We interpret the findings as consistent with both mechanisms, while remaining agnostic as to their relative contributions.

38. Online Appendix Tables H.19 and H.20 present placebo tests, showing how average cost structures, productivity, and sector aggregates change among industries not highly exposed to the minimum wage increases. Both tables show no apparent effects among these largely unexposed industries implying that the main estimates do not represent differential confounding changes in treatment relative to control states more generally, but are a result of the minimum wage policies. We also estimate extensive-margin responses and find no effect of the minimum wage on the number of active firms operating in these industries, with a point estimate of -0.0002 (std. err. $= 0.0114$).

39. Coefficients in Table V, second row, are from a regression specification similar to equation (1), but without firm fixed effects, so the estimates reflect the differential characteristics of active firms in treatment relative to control states five years after the minimum wage increases.

V. Minimum Wage Increases and Low-Income Workers

The firm-level analysis in Section IV shows that firms adjust to higher minimum wages by modestly reducing part-time hiring and largely by using new revenues to offset higher pay to low-income workers. The lack of meaningful employment effects at independent businesses raises the possibility that higher wages lead to higher incomes for the lowest-paid workers. However, depressed entry of independent businesses, reduced part-time hiring of teenagers, and potentially different adjustments at corporations will also shape the income trajectories of workers. To understand how workers are affected by wage-floor policies, we turn from our firm-level analyses to panels of individuals that include workers employed by corporations and those with periods of nonemployment.

We construct two individual-level panels to describe the trajectories of (i) low-earning workers, where low-earners are those earning less than \$20k in total individual wages working in any industry or type of firm the year before the minimum wage increase ($s-1$) and are either not working or earning less than \$25k in year $s-2$; and (ii) young people between ages 15 and 26 in the year before the minimum wage increase ($s-1$), who may be working in any kind of industry or firm or not employed. Unlike our firm analysis, these panels span employment at all types of firms, including corporations, and all industries, including those less exposed to wage-floor policies. Accordingly, the earnings and employment effects reported in this section reflect the economy-wide effects of minimum wage policies on these subsets of workers.

Each panel offers distinct insights, but also has certain limitations. The panel of low-earning workers shows us how the earnings and employment of typical low-wage workers are affected by minimum wage policies, but because it conditions on employment in the year before the policy change, it cannot capture effects on worker entry. The panel focused on young people, on the other hand, reveals how the entry and earnings of less experienced workers and nonworkers, who may lose their first foothold in the workforce as independent businesses hire fewer part-time teenagers, are affected by wage floors. Of course, a panel of young people fails to describe the experience of other minimum wage workers. Together, the two panels and the subpopulations therein provide a detailed picture of how the incomes and work opportunities of the people most likely to be affected by the policy changes evolve in the years after the minimum wage is raised.

In each regression specified in equation (2), the earnings and employment trajectories of low-earning or young individuals in control states effectively stand in for how earnings and employment may have evolved in treatment states had they not raised their minimum wage. As such, these control trajectories help account for any mean reversion or macroeconomic trends affecting the earnings of low-earning or young people similarly in control and treatment states. Earnings and employment patterns that track each other in the years before the policy change bolster the exclusion restriction—that outcomes would have been similar in treatment and control states in the absence of any policy change.

V.A. *Earnings*

Earnings are a key measure of how minimum wages affect the material well-being of low-income individuals, as they reflect the combined effects on wage rates, employment, and hours worked. Figure VII examines how higher minimum wages affect the earnings of low-earning workers and young individuals who may or may not be employed at baseline. In each panel, the estimates in the years before the minimum wage increase are precise zeroes, indicating that individuals in our treatment and control states had very similar earnings trajectories before the policy change.

Panel A plots the evolution of average earnings for low-earning workers in treated states relative to similar workers in control states. Earnings rise substantially following the minimum wage hikes, with an average increase of \$1,473 a year four years later, and average earnings gains of \$1,535 for low-earning workers initially employed in highly exposed industries. These gains are meaningful, amounting to an 18% increase for low-earning workers overall and a 15% increase for those initially employed in highly exposed industries (Online Appendix Figure I.11 and Table VI). The estimates do not condition on being employed in any year other than year $s-1$, and as such include spells of nonemployment when workers' earnings are recorded as zeroes.

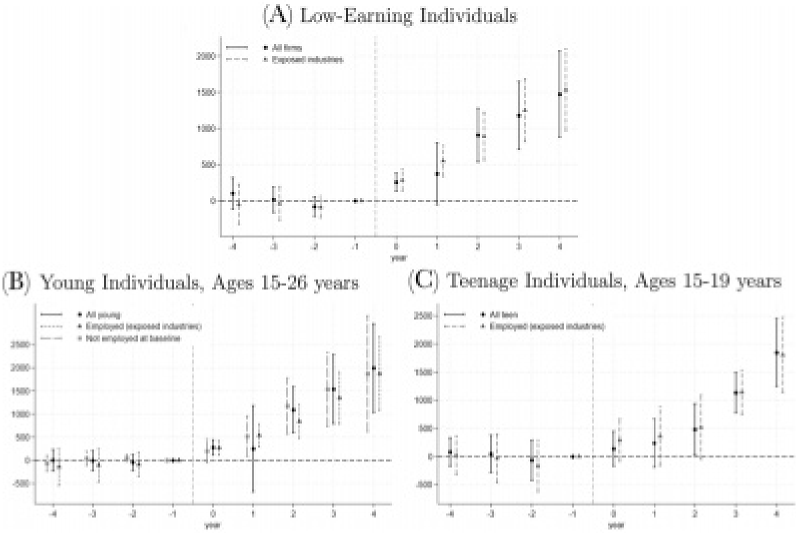

FIGURE VII

The Effect of Minimum Wage Increases on Earnings Trajectories

This figure demonstrates the effect of minimum wage increases on the evolution of average earnings for low-earning and young workers. Plotted coefficients are estimates of β_s from equation (2) where the dependent variable is annual earnings. Panel A plots the difference in the earnings trajectories of low-earning workers ($<$ \$20,000 but $>$ \$0 in year $s-1$, and $<$ \$25,000 in year $s-2$) in treatment relative to control states. Trajectories are plotted separately for all low-earning individuals and those employed in highly exposed industries in base year $s-1$. Panel B plots the effect on earnings trajectories for young individuals ages 15–26 in base year $s-1$. These individuals may or not be working at baseline and their earnings trajectories are plotted separately by base-year employment status. Panel C focuses on individuals who were teenage and may or may not have been working at baseline. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

We turn to the panel of young individuals to explore whether the reduction in the hiring of young, part-time workers at independent firms might lead minimum wage increases to actually dampen the earnings trajectories of young workers. Figure VII, Panel B tracks the earnings of our panel of young individuals overall, and by age and baseline employment status. Young workers benefit quickly and substantially, with a relative increase of \$1,995 a year four years after the reform. Panel C shows that teenagers see a similar average increase of $1,845.[40] These panels also indicate that young people not employed at baseline experience comparable gains in average earnings four years later.

V.B. Employment

Figure VIII reports the employment effects of the minimum wage hike for low-earning and young individuals. Specifically, the estimates measure the change in employment rates over the five-year period between the year before and four years after the minimum wage increases. Employment effects are flat for all groups of low-earning workers. Young individuals, on the other hand, are generally slightly more likely to be employed after the minimum wage increase, with the average baseline 15- to 26-year-old having a 1.39 percentage point higher probability of employment. Teenagers show a similar increase, though teens not employed at baseline are not more likely to be employed.[41]

V.C. Robustness

The results show that four years after the wage-floor increases were legislated, ostensibly vulnerable workers experienced substantial earnings gains on average and no decline in annual employment rates. Online Appendix Figures I.17 and I.18 show that these findings hold when focusing specifically on workers employed by highly exposed independent businesses at baseline. Online Appendix Tables I.22 and I.23 show that the controls included in regression specification (2) attenuate the quantitative estimates but do not drive the qualitative results and demonstrate that the results are very similar using "regression-adjusted" estimators of the treatment effects.

40. Again, the associated percent changes for all young workers and all teens are substantial, amounting to 21.8% and 36.9%, respectively (Online Appendix Figure I.11).

41. We estimate own-wage elasticities for these panels as the percent change in employment over the percent change in average annual earnings. For employment, the percentage point changes displayed in Figure VIII are converted to percent changes, evaluated using the observed employment rates among control firms as a counterfactual. The percent change in annual earnings is estimated using equation (2), where the outcome is log individual annual earnings. Standard errors are calculated using the delta method. The associated own-wage elasticities are −0.013 (std. err. = 0.036) for all low-earning workers, 0.025 (std. err. = 0.044) for low-earning workers in highly exposed industries, 0.064 (std. err. = 0.032) for all young individuals, and 0.038 (std. err. = 0.019) for teens.

TABLE VI

EMPLOYMENT PATTERNS OF LOW-EARNING WORKERS AND TEENAGERS

	Earnings	Employed	Exposed industry	Exposed industry independent business	Exposed industry large C-corp.	Unexposed industry independent business	Unexposed industry large C-corp.
Panel A: Low-earning workers panel							
Baseline industry							
All workers							
Exposed industries	0.149***	0.0038	0.0063*	−0.0126**	0.0173***	0.0052	−0.0047
	(0.0538)	(0.0066)	(0.0036)	(0.0055)	(0.0065)	(0.0037)	(0.0032)
Baseline (†)/control (c) mean	7,848†	0.838c	0.396c	0.686†	0.219†	0.248c	0.159c
Unexposed industries	0.172***	−0.0042	−0.0045	−0.0008	−0.0023**	0.0012	0.0031
	(0.0407)	(0.0070)	(0.0040)	(0.0024)	(0.0011)	(0.0056)	(0.0060)
Baseline (†)/control (c) mean	8,466†	0.775c	0.131c	0.063c	0.023c	0.561†	0.230†
Movers							
Exposed industries	0.191***	—	−0.0052	−0.0204**	0.0149**	0.0060	−0.0089*
	(0.0593)	—	(0.0046)	(0.0082)	(0.0071)	(0.0063)	(0.0042)
Baseline (†)/control (c) mean	7,848†	1.0c	0.322c	0.686†	0.219†	0.342c	0.207c
Unexposed industries	0.234***	—	−0.0040	−0.0004	−0.0032*	−0.0005	−0.0030
	(0.0559)	—	(0.0055)	(0.0035)	(0.0018)	(0.0111)	(0.0072)
Baseline (†)/control (c) mean	8,467†	1.0c	0.161c	0.100c	0.036c	0.561†	0.230†

TABLE VI
CONTINUED

	Earnings	Employed	Exposed industry	Exposed industry independent business	Exposed industry large C-corp.	Unexposed industry independent business	Unexposed industry large C-corp.
Panel B: Teens, repeated cross section							
All teens	−0.0117	−0.0050	−0.0094	−0.0102**	0.0028	−0.0017	−0.0030
	(0.0554)	(0.0061)	(0.0065)	(0.0040)	(0.0026)	(0.0022)	(0.0023)
Baseline mean	1,849	0.384	0.176	0.098	0.066	0.079	0.068
Employed teens	0.0684***	—	−0.0057	−0.0121*	0.0088**	−0.0006	−0.0033
	(0.0290)	—	(0.0096)	(0.0071)	(0.0044)	(0.0030)	(0.0037)
Baseline mean	4,818	1.0	0.458	0.254	0.171	0.205	0.178

Notes. The table reports employment patterns for individuals from our low-earning panel and repeated cross sections of teenagers. Employment patterns are tracked by industry and firm type. Panel A describes the effect of minimum wage increases on employment transitions between years $s-1$ and $s+4$ for a panel of low-earning workers. These workers are all employed at baseline in either exposed or unexposed industries and the estimates report the change in their transition patterns relative to similar workers in untreated states. Each row describes transition patterns conditional on the industry in which low-earning workers were employed in the base year. Movers columns focus on transition patterns of workers no longer employed by their baseline ($s-1$) employer in year $s+4$. Estimates are from a linear probability model specification of equation (2) where the dependent variable is an indicator for a worker being employed at a given type of firm in year $s+4$. The lower panel describes employment patterns for randomly sampled repeated cross sections of teenagers. Online Appendix L.4 details the construction of these data. The reported estimates convey five-year changes in average employment rates in different industries and firm types measured between years $s-1$ and $s+4$. The specifications include baseline age and age squared, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. Panel A includes individual fixed effects. All standard errors are clustered at the state-cohort level. To facilitate interpretation of magnitudes, we include baseline (year $s-1$) means when they are not 0 or 1, and control means in year $s+4$ otherwise. For the baseline low-earning workers panel (Panel A), 36% work in highly exposed industries and 64% in unexposed industries at baseline. Regression N's by row: Panel A: 987,388; 1,485,775; 842,201; 1,335,715. Panel B: 5,338,697; 2,370,428. *** $p<.01$, ** $p<.05$, * $p<.1$.

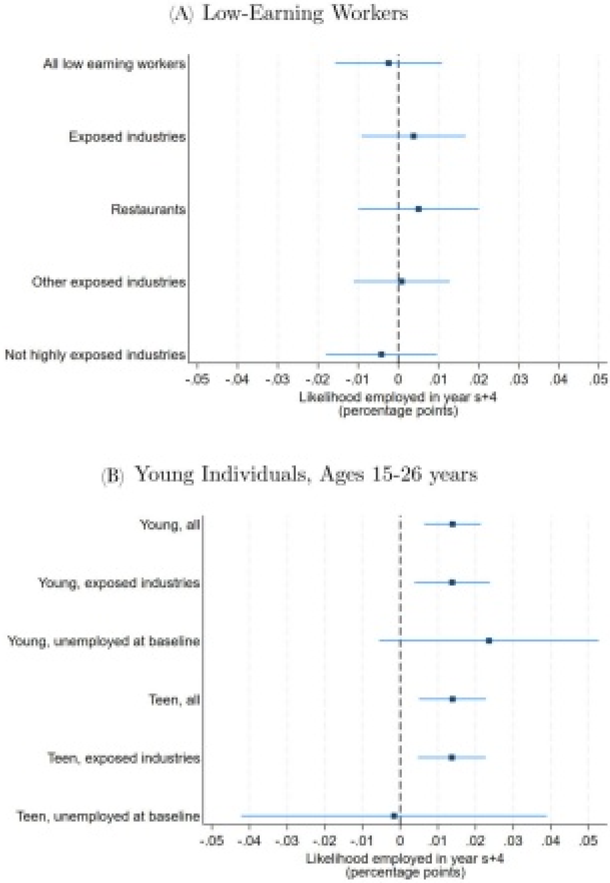

FIGURE VIII

The Effect of Minimum Wage Increases on Individual Employment

This figure demonstrates the effect of minimum wage increases on employment four years after the minimum wage increases for baseline low-earning workers and young individuals. Estimates are from linear probability model regressions of equation (2) where the dependent variable is an indicator if the individual receives a W-2 in year s. Plotted coefficients are estimates of the average change in employment probability for each group between the base year prior to the policy change $s-1$ and four years hence $s+4$. Panel A plots the employment effects on low-earning workers overall and for subgroups divided by baseline employer type. Panel B describes employment effects for young individuals ages 15–26 in base year $s-1$. These individuals may or not be working at baseline, and estimates are reported separately by baseline employment and age. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

We also conduct placebo tests where we estimate differences in worker outcomes between treatment and control states around a placebo minimum wage increase in 2010 (i.e. at least three years before the first actual increase). Similar to Dustmann et al. (2022) and Paul-Delvaux (2024), this exercise helps establish that the observed differences in worker outcomes after the minimum wage increases stem from the policy changes, not from underlying differences in worker outcomes across states. Reassuringly, individual earnings paths are very similar in treatment and control states before and after the placebo minimum wage increases, as are employment outcomes (Online Appendix Figures I.13 and I.14).

V.D. *Retention*

In addition to earnings and employment, minimum wage increases may affect where low-earning workers work. We start with retention rates, defined as the probability that a worker remains employed by their baseline firm after the minimum wage increases. Figure IX plots estimates of the differential five-year retention rate for all low-earning workers and subsets conditional on baseline employer type.[42] Retention rates increase significantly for low-earning workers in states that raise the minimum wage, by 1.30 percentage points. Retention rates rise similarly for low-earning workers employed by independent businesses and large corporations in highly exposed industries. These findings mirror the turnover reductions previously demonstrated for specific employers (Reich, Hall, and Jacobs 2005; Coviello, Deserranno, and Persico 2022) or by worker rather than firm characteristics (Dube, Lester, and Reich 2016). In contrast, retention does not rise in unexposed industries that employ small shares of minimum wage workers.

Higher retention can benefit firms through direct savings from lower search, recruitment, and training costs and indirect gains from a more experienced and efficient workforce (Jäger, Heining, and Lazarus forthcoming). Changes in retention rates also inform the interpretation of estimated firm-level employment responses. The tax data report the number of W-2s issued by a

42. To estimate differential retention rates, we estimate equation (2) using a linear probability model (LPM) specification where the outcome is an indicator for working at the same firm in year s as in year $s-1$. We focus on the low-earning panel because measuring retention requires employment in the base year.

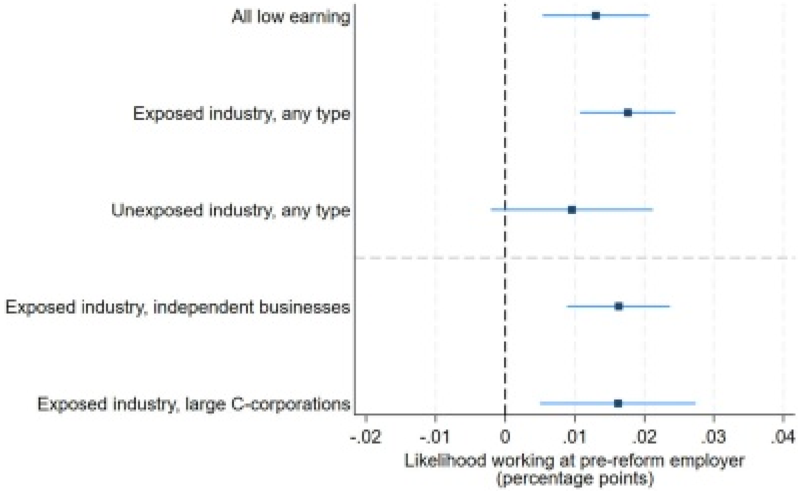

FIGURE IX

The Effect of Minimum Wage Increases on Worker Retention

This figure demonstrates the effect of minimum wage increases on retention rates for low-earning workers. Low-earning workers are those employed at baseline but earning < \$20k and also either not working or earning < \$25k the year before, in $s-2$. Plotted coefficients are from linear probability model regressions of equation (1) where the dependent variable is an indicator equal to one if a worker has the same employer in year $s+4$ as at baseline $s-1$. Estimates are reported separately by industry and type of employer at baseline. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density, and quintiles of county employment rates—all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

firm to distinct workers in a given year, capturing a firm's total number of employment relationships rather than the number of jobs at a firm at a given time. Let us denote this employment concept E_{jt}.

We can define the average number of "jobs" in a firm in year t as N_{jt}.[43] Let $n_{jt} \geqslant 1$ denote the average number of workers needed to fill a job in a year. If all positions at a firm are filled by one worker for the entire year, then $n_{jt} = 1$; if there is worker turnover for some positions within a year, $n_{jt} > 1$.

43. A job can be defined arbitrarily and the number and composition of jobs will vary across firms. That is, N_{jt} could represent some number of full-time and/or part-time positions that a firm needs to fill to operate at its given production level.

Then, $E_{jt} = N_{jt} n_{jt}$ and $\ln(E_{jt}) = \ln(N_{jt}) + \ln(n_{jt})$. An estimated log change in employment relationships therefore is the sum of the log change in jobs and the log change in workers needed to fill each job. As such, an estimated reduction in employment relationships may reflect a reduction in the number of jobs and/or a reduction in turnover (a decrease in n_{jt}).

Higher retention implies that the estimated reduction in E_{jt} reported in Section IV likely overstates the reduction in jobs from a firm perspective. If we assume that one less separation offsets the need for one other worker to fill that job, then the estimated 1.87% reduction in separations[44] would imply that the estimated 2% reduction in employment relationships corresponds to only a 0.13% reduction in jobs. Without information on hours or tasks, we cannot directly estimate the effect on n_{jt}. As the job losses we observe amount to roughly a quarter of a year of full-time work at the minimum wage, and retention rises over multiple years, it is plausible that each retained worker supplants at least one missing employment relationship such that there is no meaningful reduction in jobs at highly exposed independent businesses.

V.E. Worker Transitions

In addition to boosting worker retention, minimum wage increases affect how workers move between employers and industries. The upper panel of Table VI compares low-earning worker transitions in states that raised their minimum wages to transitions in control states.[45]

Low-earning workers who leave highly exposed firms ("movers") are significantly less likely to work at independent firms in highly exposed industries four years after a minimum wage increase, but are substantially more likely to work in large C-corporations in these industries. Online Appendix Table K.25 shows that correspondingly, workers are more likely to move to large firms in highly exposed industries and somewhat less likely

44. The five-year separation rate for workers in highly exposed industries in independent businesses in control states is 86.9%, so the 1.63 percentage point estimate is associated with an estimated 1.87% reduction in reduction in separation rates in treatment states.

45. See Online Appendix L for details on classifying employer types in the tax data.

to move to small firms. These patterns are consistent with a reduction in available positions at independent businesses due to the reduced entry of smaller, less productive firms and resulting reallocation toward larger firms. These offsetting effects, along with higher retention, leave low-earning workers no less likely to remain employed in highly exposed industries.[46]

This direct evidence of worker reallocation in response to increases in the minimum wage complements the findings of Dustmann et al. (2022). We also highlight another channel through which the minimum wage affects worker flows by showing the increase in retention rates among independent firms and C-corporations in highly exposed industries.

V.F. Reconciling Firm- and Individual-Level Employment Effects

The transitions reported in Table VI help explain how the missing employment relationships at highly exposed independent firms documented in Section IV do not translate into negative employment effects at the individual level. The observed reallocation patterns indicate that the decline in employment at independent businesses is largely offset by worker movements to large C-corporations in highly exposed industries. These transitions across firm types broadly reconcile the reductions in employment opportunities at independent businesses with the stable overall employment rates observed at the worker level.

Our analyses can further contextualize the relationship between the observed effects on independent businesses and the employment patterns of individual workers. We estimate a 3.3%

46. Reference means are included to facilitate interpretation of the magnitudes of the estimates. For example, the estimated 1.26 percentage point reduction in employment in highly exposed independent businesses evaluated at the baseline sample mean of these workers implies a 1.8% reduction in employment among these workers. This estimate is very similar to the 2% employment reduction we estimate in the firm-level analysis of highly exposed independent businesses (Online Appendix Table H.17). This shows that estimated employment effects among these firms are similar using the firm-level and individual-level specifications, equations (1) and (2), respectively. Online Appendix Table K.26 shows the differential two-year transition rates. Retention effects are decreasing over time, as we would likely expect given the high turnover in these industries and among low-wage workers. Transitions from smaller (independent) firms in highly exposed industries to larger firms (C-corporations) happen fairly quickly, generally having already materialized by year $s+2$.

aggregate reduction in employment relationships among low-earning workers at independent firms. Of these, 2% are from reduced employment among incumbent firms and 1.3% result from reduced firm entry. Of the 2% reduction among incumbents, roughly three-quarters (1.5%) is attributable to reduced hiring among part-time teenagers.[47]

Table VI shows that following the minimum wage increases, low-earning workers are less likely to work at highly exposed independent firms. The estimates imply a 1.5% decline in employment relationships between low-earning workers and independent firms,[48] which corresponds closely with the 1.3% decline in employment relationships due to reduced entry of independent firms into highly exposed industries plus the 0.5% reduction in non-teen hiring at incumbent firms. Effectively, as workers churn in highly exposed industries (albeit at a reduced rate due to higher retention), they are more likely to rematch with a C-corporation rather than an independent business as reduced entry curtails the number of independent firms.

Reduced hiring among incumbent firms primarily affects teenagers. It is important to note that the hiring reductions we document at the firm level pertain to contemporaneous teenagers—those who are ages 15 to 19 in year $s+4$—while our teen panel follows those who were ages 15 to 19 in year $s-1$.

47. Based on the aggregate analysis using collapsed data described in Section IV.F, reduced hiring at incumbent firms is associated with a 2% net reduction in employment, while reduced firm entry leads to a 4.6% overall decline in employment relationships. On average, 28% of employees at highly exposed independent businesses are low-earning workers, so we estimate that $4.6\% \times 28\% = 1.3\%$ of losses from lower entry are applicable low-earning workers. From Section IV.A, we know 75% of the 2% reduction in employment relationships among incumbent firms, or 1.5%, are missing hires of teenagers.

48. The estimated 1.5% reduction in employment relationships is a weighted average of (i) the implied 1.8% decline in the probability that low-earning workers previously employed in highly exposed industries work at highly exposed independent businesses ($\frac{-0.0126}{0.686} = 0.0184$) and (ii) the implied percent decline in the probability that low-earning workers previously employed in unexposed industries work at highly exposed independent businesses ($\frac{-0.0008}{0.063} = 0.013$). Because 36% of low-earning workers in our sample are employed by highly exposed industries at baseline, the weighted average yields a 1.5% decline in employment relationships between low-earning workers and highly exposed independent firms ($0.36 \times 1.84 + 0.64 \times 1.3 = 1.50\%$).

As a result, our teen panel estimates also capture the effects of these individuals aging over the analysis period. To measure contemporaneous teen employment changes between years $s-1$ and $s+4$, we complement our analysis with a repeated cross section of teenagers in treated and control states.[49] The difference-in-differences estimates of the effect of the minimum wage increases in these repeated cross sections are reported in the lower panel of Table VI.[50]

The analysis shows that teenagers are not significantly less likely to be employed after the minimum wage increases. However, employed teens are significantly less likely to work in independent firms in highly exposed industries and are significantly more likely to work at large C-corporations in these industries. The results suggest that teens have roughly 1.3% fewer employer relationships at highly exposed independent businesses after the minimum wage increase, which nearly matches the 1.5% reduction in teen hiring we estimate in the firm-level data.[51] We also find evidence of less multiple job holding as working teens have 2.8% fewer employment relationships in a given year, which likely further buffers estimated employment effects at the firm-level.[52]

In Online Appendix N, we validate these findings using public data from the American Community Survey (ACS; Ruggles et al. 2024). While the tax data are not designed to estimate employment rates per se, the ACS is. The tax data have the advantage of linking workers and firms over time, allowing us to estimate the dynamics across firm types that underlie aggregate patterns,

49. Details of the repeated cross section are in Online Appendix L.4.

50. The earnings effects for contemporaneous teens are smaller than the panel estimates, indicating that aging contributes to the latter. Also, since most teenagers are not employed in a given year, the average effects are attenuated (first row), while employed teenagers still see significant earnings increases (second row).

51. The cross-sectional analysis shows that teenagers (whether working or not at baseline or in any year) have a 1 percentage point, or 7.25%, lower probability of working at highly exposed independent businesses. Because teenagers account for roughly 18% of workers in highly exposed independent firms, the 7.25% reduction in teen workers corresponds to a 1.3% decline in total employment relationships at highly exposed firms.

52. We estimate that teens with any employment in treatment states hold 0.045 (std. err. = 0.015) fewer jobs in $s+4$, evaluated at the average and median number of jobs held of 1.6.

whereas the ACS provides a representative cross section of the population regardless of employment status and enables direct identification of minimum wage workers. The ACS data confirm the main findings from the tax data: negligible individual employment effects across worker types and in highly exposed industries, and wage increases similar to those observed in our low-earning worker panel and repeated cross sections of teenagers.

VI. Discussion

Our empirical results show that incumbent independent businesses accommodate minimum wage increases without laying off workers or experiencing profit declines. Instead, revenues rise enough to fully offset the added labor costs such that consumers rather than owners bear the burden of higher wage floors. Minimum wages deter entry, such that firms that enter despite the added costs are positively selected with higher value added per worker and lower nonlabor costs. Incumbent firms, too, become more productive and efficient after the minimum wage increase. At the aggregate level, summing across all independent businesses in highly exposed industries, workers receive a larger share of expanded revenues generated by fewer, more productive firms whose profits remain unchanged.

Online Appendix O presents a model of imperfect product market competition with fixed costs and heterogeneous production technologies to highlight how extensive-margin responses and heterogeneity can mediate the effects of minimum wage increases. In our Cournot framework, reduced entry reallocates revenues toward the more productive and efficient firms that continue to enter and operate in these highly exposed industries despite higher wage floors, facilitating firm revenue increases large enough to mute profit effects. This exercise is in the style of Besley (1989), who shows that when firms can exit under Cournot competition, a specific (per unit) tax on output can increase output per firm and total welfare depending on the shape of market demand. Unlike a specific tax, minimum wages raise the price of labor inputs, so the firm's cost shock will depend directly on its production function. Firms whose production technologies rely more or less heavily on minimum wage labor, or use minimum wage labor more or less efficiently, will experience different cost shocks even when producing the same product. We incorporate this conceptually and empirically relevant heterogeneity in our model.

Under Cournot competition, market shares are proportional to margins, with the most efficient firms enjoying the largest margins and market shares. Even absent entry effects, a cost shock reallocates demand toward ex ante more efficient or less shocked firms. Reduced entry further amplifies this reallocation, shifting demand from firms that do not enter to operating firms, thereby facilitating revenue increases for each firm and dampening the profit effects.

Our estimates show that the minimum wage increases reduce firm entry by 2%, consistent with narrower margins leaving some firms unable to cover fixed costs. Entry declines among lower-productivity and less efficient firms, meaning that entrants are positively selected. We also observe productivity and efficiency gains among incumbents. This smaller set of more productive firms experiences average revenue growth of 4.3% (std. err. = 0.0048), with particularly large gains among firms with higher baseline value added (Online Appendix Table F.10). Revenue reallocation combined with productivity gains helps explain how independent businesses maintain their profits. The broad reach of the cost shock also facilitates price increases. Although firms differ in their reliance on low-wage labor, most firms in highly exposed industries employ some low-wage workers. In contrast to a firm unilaterally choosing to raise prices, minimum wage increases raise costs industry-wide, and as such the elasticity mediating price increases is the (lower) elasticity of market rather than firm demand, easing the pass-through of cost increases into prices. Moreover, if customers attribute price increases to regulations rather than firm behavior, reduced antagonism may further temper demand responses.

Our framework considers how imperfect product market competition can rationalize our empirical findings. Models of imperfect competition in labor markets also predict reallocation in response to cost shocks. Bhaskar and To (1999) show that under monopsony with free entry, minimum wage hikes can raise employment per firm and induce firm exit, with ambiguous effects on total industry employment and welfare. Building on Butcher, Dickens, and Manning (2012) and like Bassier (2023), Dustmann et al. (2022) show that worker reallocation from smaller, less productive, lower-paying firms to larger, more productive, higher-paying firms is consistent with a model of monopsony power in the labor market and heterogeneous production costs. Higher wage floors reallocate workers to more productive firms as the least efficient firms exit and employment expands among surviving firms where the minimum wage binds. Consistent with these models, we find that low-earning workers and teens reallocate from independent businesses to C-corporations (Table VI), and that lower value-added firms account for the decline in employment relationships (Online Appendix Table F.10). Like these models, our findings and framework underscore the roles of firm heterogeneity, market imperfections, and extensive-margin dynamics in shaping the effects of the minimum wage.

VII. Conclusion

The twenty-first century has seen renewed global interest in minimum wages as a policy tool for intervening in low-wage labor markets and addressing rising inequality. Countries such as the United Kingdom and Germany have reintroduced minimum wages, while others such as Hungary and Brazil have significantly raised their wage floors (Dube and Lindner 2024). In the United States, 30 states have raised their minimum wages since 2015, reflecting the broad popularity of using price floors to boost the wages of low-skilled workers.

While an extensive literature demonstrates that minimum wage increases have minimal employment effects in the short to medium term, there remains limited understanding of how markets adjust to accommodate these wage hikes. Our study sheds light on how independent firms in the United States respond to minimum wage increases. Contrary to concerns that such wage increases might imperil small firms, our comprehensive analysis reveals that independent firms demonstrate remarkable adaptability. Despite wage pressures, these businesses do not lay off existing workers; instead, they pare back part-time hiring as worker retention rises. They mainly accommodate the higher labor costs through added revenues and are successful enough in generating new sales that profits are unchanged.

Our findings also highlight sector-wide implications, where minimum wage increases modestly reduce new firm entry, favoring more productive and efficient entrants. Among incumbent firms, value added per worker rises on average, and revenues grow more among firms with higher baseline value added. As such, minimum wages reshape industries that rely on low-wage workers, such that they feature fewer but more productive firms after the cost shock. Through changes in entry and demand reallocation, the minimum wage shifts the productivity distribution of independent firms in a manner akin to how the emergence of new technologies (Collard-Wexler and De Loecker 2015), exposure to international trade (Melitz 2003), and recessions (Osotimehin and Pappadà 2017) can alter the productivity distribution in an industry.

At the individual level, low-earning and young individuals experience significant earnings gains and stable employment after minimum wage increases. There are also worker transitions toward larger C-corporations. These shifts illustrate how worker mobility can mitigate potential employment effects as rising retention and reduced part-time hiring shrink job openings at independent firms.

It is important to note that our findings center on the short- to medium-run effect of phased-in minimum wage increases. The longer-run effects may differ if new entrants eventually adopt production methods that rely less on low-wage labor, or if incumbents reconfigure their inputs away from the workers most affected by these policies. It is also possible that firms respond to the increased labor costs along margins not estimated here, such as reducing workplace amenities or deferring equipment maintenance.

Supplementary Material

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

Data Availability

The code underlying this article is available in the Harvard Dataverse, https://doi.org/10.7910/DVN/3ZXSCE (Rao and Risch 2025).

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